

Network-Optimal Single-Loop Decentralized Bilevel Optimization via Chebyshev-Optimistic Gradient Tracking

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We study deterministic decentralized bilevel optimization over a connected communication network. Each agent i privately stores local upper- and lower-level functions f_i and g_i , and the network seeks to minimize the global bilevel value

$$F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}, \mathbf{y}^*(\mathbf{x})), \quad \mathbf{y}^*(\mathbf{x}) \in \operatorname{argmin}_y \frac{1}{n} \sum_{i=1}^n g_i(\mathbf{x}, y)$$

using only neighbor-to-neighbor communication and local first- and second-order-vector oracle access. The main difficulty is that the hypergradient of F couples three global objects: the outer variable $\mathbf{x}$, the lower solution $\mathbf{y}^*(\mathbf{x})$, and the adjoint solution of a global lower-level linear system. Existing decentralized bilevel methods either pay nested communication costs, require strong heterogeneity assumptions, or do not isolate the separate roles of network mixing, outer conditioning, and inner-adjoint conditioning.

We propose B-CO-AGT, a single-loop decentralized method that combines Chebyshev-accelerated consensus, optimistic acceleration on the outer variable, and simultaneous gradient tracking for the lower-level residual, adjoint residual, and hypergradient estimator. In the strongly-convex-strongly-convex regime, we prove a Lyapunov contraction and obtain communication complexity $\tilde{\mathcal{O}}(\sqrt{\chi}\sqrt{\kappa_x} \operatorname{poly}(\kappa_y) \log \frac{1}{\epsilon})$ where $\chi = (1 - \rho)^{-1}$ is the network condition number, $\kappa_x = L_F/\mu_x$ is the outer condition number, and κ_y is the lower-level condition number. We complement the upper bound with independent lower bounds showing that any deterministic decentralized bilevel method must pay $\Omega(\sqrt{\chi\kappa_x} \log \frac{1}{\epsilon})$ communication rounds on network/outer-hard instances and $\Omega(\sqrt{\kappa_y} \log \frac{1}{\epsilon})$ lower linear-system oracle calls on inner-adjoint-hard instances. Thus the proposed method is network-optimal up to logarithmic and bilevel-conditioning factors, and the lower-bound construction clarifies why decentralized bilevel optimization has two independent hardness mechanisms rather than a single monolithic complexity parameter. We also discuss a convex-strongly-convex extension. In that regime, accelerated $\mathcal{O}(1/K^2)$ outer convergence follows under a weighted hypergradient-perturbation budget; a horizon-calibrated variant enforces this budget and yields a provable accelerated communication bound, while the pure one-step single-loop proof remains conditional.

CCS Concepts: • **Computing methodologies** → **Distributed algorithms**; **Machine learning algorithms**; **Multi-agent systems**; • **Theory of computation** → **Mathematical optimization**.

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1 Introduction

Large-scale learning and decision systems increasingly operate without a central coordinator. Federated and peer-to-peer systems distribute data, computation, and ownership across agents that can communicate only through a sparse graph. At the same time, many modern learning problems are naturally bilevel: hyperparameter optimization, meta-learning, data valuation, personalized models,

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adversarial training, and incentive-aware control all involve an outer objective whose value depends on the solution of an inner optimization problem.

These two trends lead to *decentralized bilevel optimization*. In its simplest deterministic form, n agents connected by a graph seek to solve

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}, \mathbf{y}^*(\mathbf{x})), \quad (1)$$

where the lower-level response is the global solution

$$\mathbf{y}^*(\mathbf{x}) := \operatorname{argmin}_{\mathbf{y} \in \mathbb{R}^m} G(\mathbf{x}, \mathbf{y}), \quad G(\mathbf{x}, \mathbf{y}) := \frac{1}{n} \sum_{i=1}^n g_i(\mathbf{x}, \mathbf{y}). \quad (2)$$

The agents do not have access to F , G , $\mathbf{y}^*(\mathbf{x})$, or the exact hypergradient. They only have access to their local functions $\{f_i, g_i\}$ and can exchange information with neighbors. The primary obstacle is that a bilevel hypergradient is not an ordinary gradient. Under lower-level strong convexity, the implicit-function formula gives

$$\nabla F(\mathbf{x}) = \nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) - \nabla_{\mathbf{xy}}^2 G(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) \mathbf{v}^*(\mathbf{x}), \quad (3)$$

where $f(\mathbf{x}, \mathbf{y}) := \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}, \mathbf{y})$, and the adjoint variable $\mathbf{v}(\mathbf{x})$ solves $\nabla_{\mathbf{yy}}^2 G(\mathbf{x}, \mathbf{y}^*(\mathbf{x})) \mathbf{v}^*(\mathbf{x}) = -\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}^*(\mathbf{x}))$. Thus computing even one exact hypergradient requires solving two global subproblems: the lower-level minimization and the adjoint linear system. In a decentralized network, both subproblems are further entangled with consensus error and gradient-tracking error.

A naive solution would be nested: run many rounds of decentralized lower-level optimization to approximate $\mathbf{y}^*(\mathbf{x})$, run many more rounds to approximate the adjoint $\mathbf{v}^*(\mathbf{x})$, then update $\mathbf{x}$, and repeat. Such methods are unattractive for communication-constrained systems. They hide the true cost behind inner loops and often over-communicate relative to the information-theoretic limit.

The central question of this paper is:

Can decentralized bilevel optimization be solved in a genuinely single loop, while matching the optimal network dependence of accelerated decentralized single-level optimization?

We answer this question affirmatively for the strongly-convex–strongly-convex regime, up to the remaining polynomial dependence on the lower-level condition number. Our algorithm, B-CO-AGT , updates all variables once per iteration: the outer variable $\mathbf{x}$, the lower variable $\mathbf{y}$, the adjoint variable $\mathbf{v}$, and three gradient trackers. It uses Chebyshev-filtered mixing to obtain accelerated network averaging, an optimistic outer step to obtain accelerated outer contraction, and gradient tracking to keep local estimates aligned with global bilevel residuals.

The resulting theory gives a clean decomposition of the complexity:

$$\text{communication} \approx \underbrace{\sqrt{\chi}}_{\text{network}} \cdot \underbrace{\sqrt{\kappa_{\mathbf{x}}}}_{\text{outer geometry}} \cdot \underbrace{\text{poly}(\kappa_{\mathbf{y}})}_{\text{inner/adjoint tracking}} \cdot \log(1/\epsilon).$$

The lower bounds show that the first two factors are unavoidable, and that the inner-adjoint system creates a separate $\sqrt{\kappa_{\mathbf{y}}} \log(1/\epsilon)$ oracle barrier even when the network is trivial. Thus, we establish that the decentralized bilevel optimization inherits the accelerated network/outer lower bound of distributed strongly convex optimization and simultaneously inherits the Krylov-type lower bound of solving the lower-level adjoint system.

Our Contributions

The contributions of this work can be summarized as follows:

- (1) **A single-loop decentralized bilevel algorithm:** We propose B-CO-AGT: Bilevel Chebyshev-Optimistic Accelerated Gradient Tracking. Each iteration performs: (a) an optimistic outer update for X_k , (b) one lower-level tracking step for Y_k , (c) one adjoint tracking step for V_k , (d) Chebyshev-filtered consensus on all updated quantities, (e) gradient tracking for the lower residual, adjoint residual, and hypergradient estimator.

No exact lower solve, exact inverse Hessian, or exact hypergradient oracle is used.

- (2) **A Lyapunov contraction for SC-SC decentralized bilevel optimization:** Under standard smoothness, lower-level strong convexity, outer strong convexity, and network assumptions, we construct a weighted Lyapunov functional coupling thirteen error modes: outer consensus, lower tracking, adjoint tracking, hypergradient bias, lagged hypergradient bias, squared spatial errors, and gradient-tracker errors. We prove that for appropriate stepsizes and Chebyshev degree,

$$\mathcal{V}_{k+1} \leq (1 - c\tau)\mathcal{V}_k, \quad \tau = \Theta(\sqrt{\mu_x \alpha_x}).$$

This yields a linear convergence guarantee.

- (3) **Network-optimal communication dependence:** With Chebyshev degree $T = \mathcal{O}\left(\sqrt{\chi} \log \frac{\mathcal{P}(\kappa_x, \kappa_y)}{\tau}\right)$,

B-CO-AGT achieves $N_{\text{comm}} = \tilde{\mathcal{O}}\left(\sqrt{\chi} \sqrt{\kappa_x} \text{poly}(\kappa_y) \log \frac{1}{\epsilon}\right)$ communication rounds in the SC-SC regime. The dependence $\sqrt{\chi} \sqrt{\kappa_x}$ matches the accelerated decentralized single-level benchmark.

- (4) **Independent lower bounds:** We establish two independent lower bounds for deterministic decentralized bilevel methods using local gossip and first-order/Hessian-vector oracle access: $\Omega(\sqrt{\chi \kappa_x} \log \frac{1}{\epsilon})$ communication rounds are necessary on a subclass where the lower problem is structurally trivial but the outer problem is hard, and $\Omega\left(\sqrt{\kappa_y} \log \frac{1}{\epsilon}\right)$ lower linear-system oracle calls are necessary on a subclass where the network is trivial but the adjoint system is hard. These lower bounds show that decentralized bilevel optimization has two independent hardness modes: network/outer hardness and lower-adjoint hardness. A multiplicative communication-HVP lower bound would require a stronger synchronized oracle model; we do not claim such a lower bound here.

- (5) **Convex-strongly-convex extension:** We extend the framework to convex-strongly-convex bilevel optimization. We prove that accelerated $\mathcal{O}(1/K^2)$ convergence follows if the hypergradient tracking error satisfies a finite-horizon weighted budget. A horizon-calibrated variant enforces this budget and achieves $\tilde{\mathcal{O}}\left(\sqrt{\chi} \sqrt{\kappa_y} \sqrt{\frac{L_F R^2}{\epsilon}}\right)$ communication with accelerated lower/adjoint correction. The pure one-step single-loop C-SC statement remains conditional unless the weighted tracking-budget proof is completed.

2 Related Work

Bilevel optimization and implicit hypergradients. Bilevel optimization has become a standard formulation for problems in hyperparameter optimization, meta-learning, data reweighting, and learning with equilibrium or response models. In the smooth setting with a strongly convex lower-level problem, the lower solution map is single-valued and the outer gradient can be expressed through an implicit-differentiation formula involving an adjoint linear system. This observation underlies much of the modern algorithmic literature on bilevel optimization, including approximation-based methods, iterative differentiation, approximate implicit differentiation, and single-timescale stochastic algorithms [1, 2, 5, 6, 8]. Recent work has also developed lower bounds and accelerated methods for

centralized bilevel optimization, showing that the cost of estimating the lower solution and the adjoint variable is not merely an artifact of analysis but an intrinsic part of the problem complexity [7].

Our work is closest in spirit to this implicit-hypergradient line, but the decentralized setting introduces a qualitatively different difficulty: the lower solution, the adjoint variable, and the hypergradient are all global quantities formed from agent-local functions. Consequently, one must control not only lower-level and adjoint approximation errors, but also consensus error, tracking error, and the interaction of these errors with outer acceleration.

Decentralized optimization and gradient tracking. Decentralized optimization has a long history in distributed averaging, consensus optimization, and peer-to-peer learning [4, 12]. A central challenge is that local gradients are biased estimators of the global gradient when data are heterogeneous across agents. Methods such as EXTRA, exact diffusion, and gradient tracking overcome this issue by combining local descent with dynamic correction mechanisms that track the global gradient [11, 16, 18, 21]. For smooth strongly convex objectives, the optimal communication complexity over a network is known to scale as $\tilde{\mathcal{O}}(\sqrt{\chi\kappa}\log(1/\epsilon))$, where χ is the network condition number and κ is the objective condition number [17]. Chebyshev acceleration is a key tool for achieving the optimal $\sqrt{\chi}$ dependence by replacing repeated raw gossip with polynomially accelerated mixing [16, 17].

The present paper can be viewed as extending this optimal decentralized optimization principle to bilevel problems. However, standard gradient tracking is insufficient: a bilevel hypergradient depends on a lower-level state and an adjoint state, both of which must themselves be tracked over the network. This leads to a coupled system of consensus, lower-residual, adjoint-residual, and hypergradient-tracking errors. Our algorithm uses three coordinated trackers rather than a single gradient tracker, and the analysis requires a Lyapunov function that couples these tracking errors with the accelerated outer dynamics.

Decentralized bilevel optimization. Several recent works have studied bilevel optimization over decentralized or federated networks. Existing methods have addressed stochastic nonconvex–strongly-convex settings, decentralized hypergradient estimation, heterogeneity correction, and communication-efficient derivative evaluation [3, 14, 19, 20, 22, 23]. A particularly relevant line of work develops single-loop decentralized bilevel algorithms, where the lower-level and upper-level variables are updated simultaneously rather than through expensive nested inner loops [3, 23]. Other recent methods reduce the cost of Hessian-vector or Jacobian-vector evaluations through snapshotting, variance reduction, or first-order reformulations [19, 20]. Work beyond the strongly convex lower-level regime has also begun to appear, for example by replacing strong convexity with weaker structural assumptions or alternative stationarity measures [14].

Our contribution is different in emphasis. Rather than focusing only on stochastic oracle complexity or practical derivative reduction, we study the deterministic decentralized strongly-convex–strongly-convex and convex–strongly-convex regimes from a rate-optimality perspective. The goal is to identify the correct dependence on the network condition number, the outer condition number, and the lower-level condition number. The proposed B-CO-AGT method combines Chebyshev-accelerated communication, optimistic outer acceleration, and simultaneous tracking of the lower residual, adjoint residual, and hypergradient. This yields an upper bound with accelerated network dependence $\sqrt{\chi}$, matching the optimal dependence known for single-level decentralized optimization.

Lower bounds and rate optimality. Lower bounds are essential for understanding which parts of a decentralized bilevel rate are algorithmic artifacts and which are unavoidable. For single-level decentralized strongly convex optimization, the lower bound of Scaman et al. [17] implies that any first-order decentralized method must pay at least $\Omega(\sqrt{\chi\kappa_x}\log(1/\epsilon))$ communication rounds in the worst case. Centralized bilevel lower bounds show that even without a network, approximating the